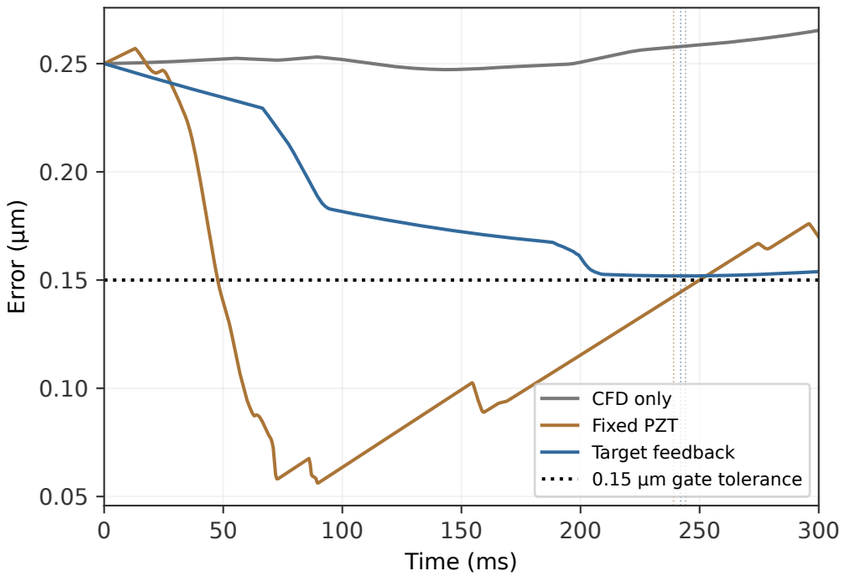
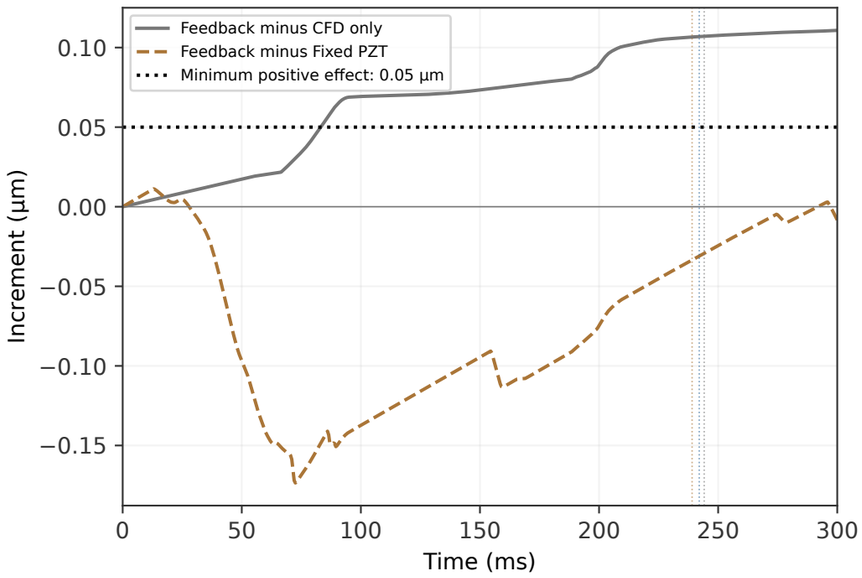
Control events, transverse motion and numerical quality
Vertical dotted lines: each condition crosses the same downstream gate

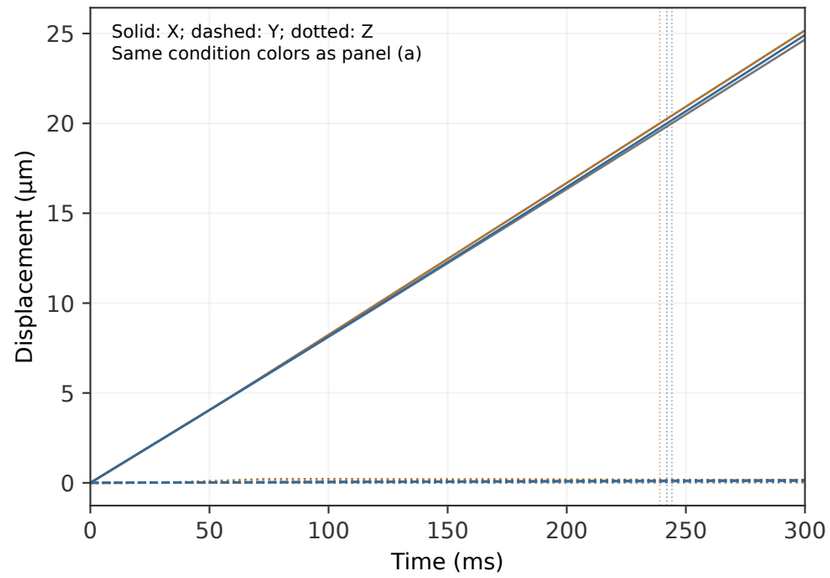
(a) Transverse target error



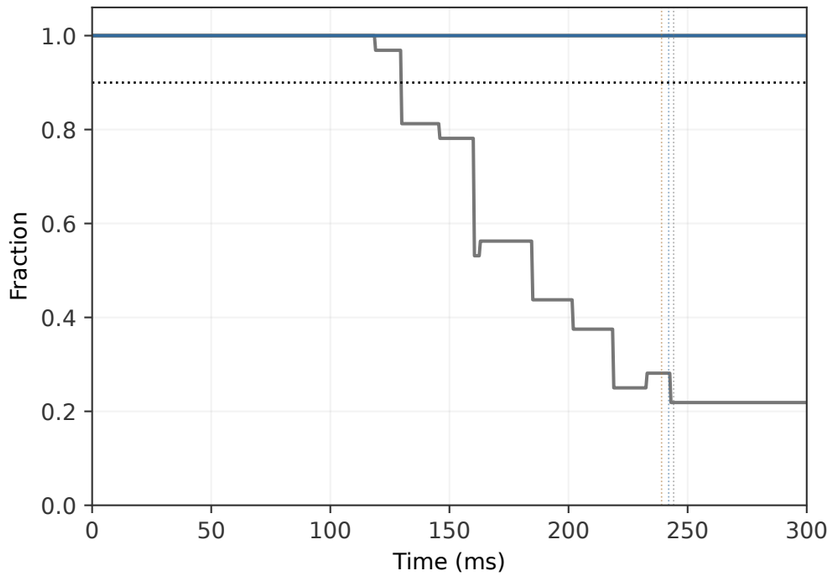
(b) Same-time +Z control increments



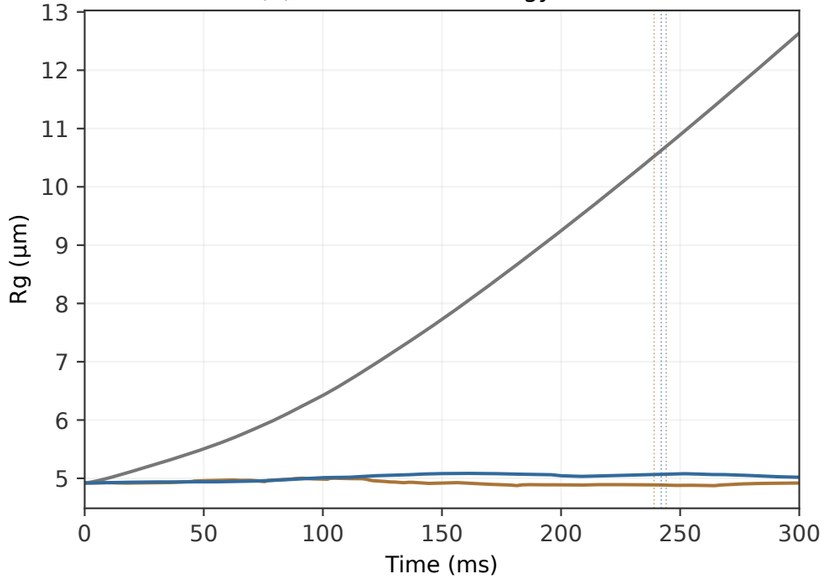
(c) Fixed-ID displacement



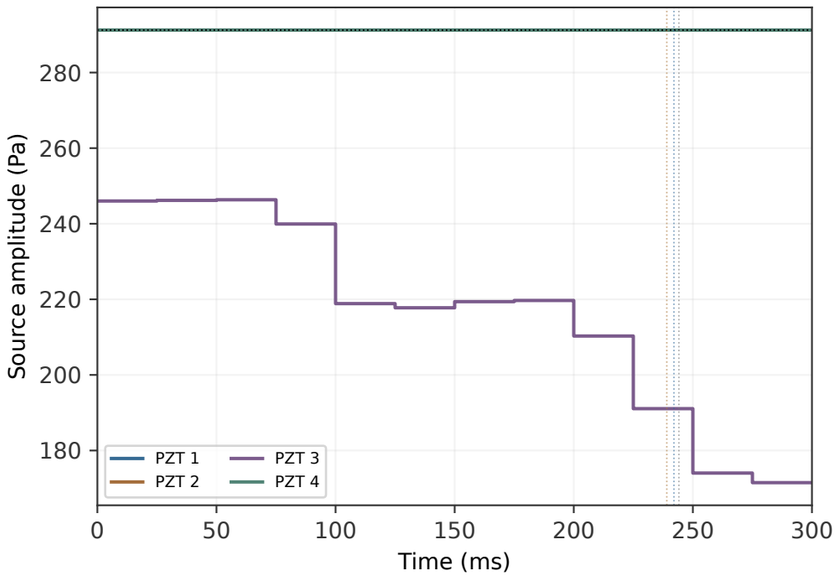
(d) Largest connected fraction



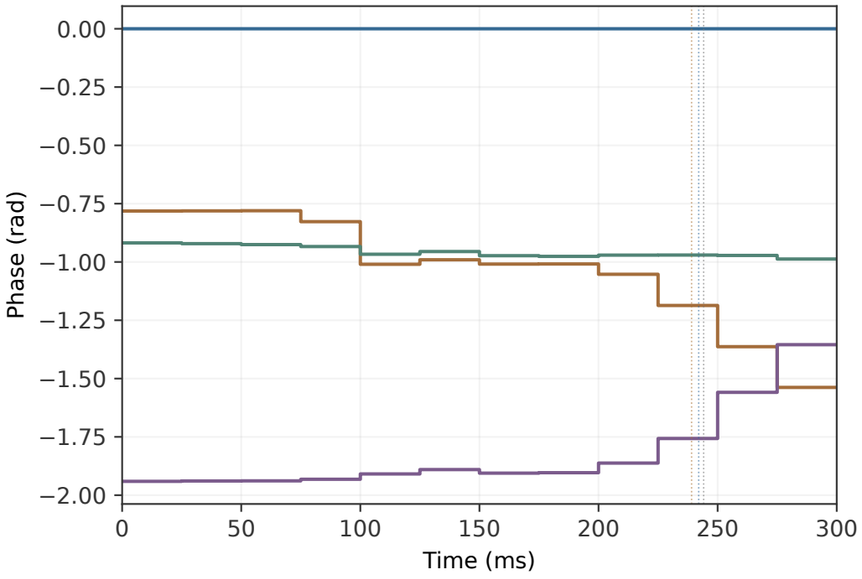
(e) Fixed-ID radius of gyration



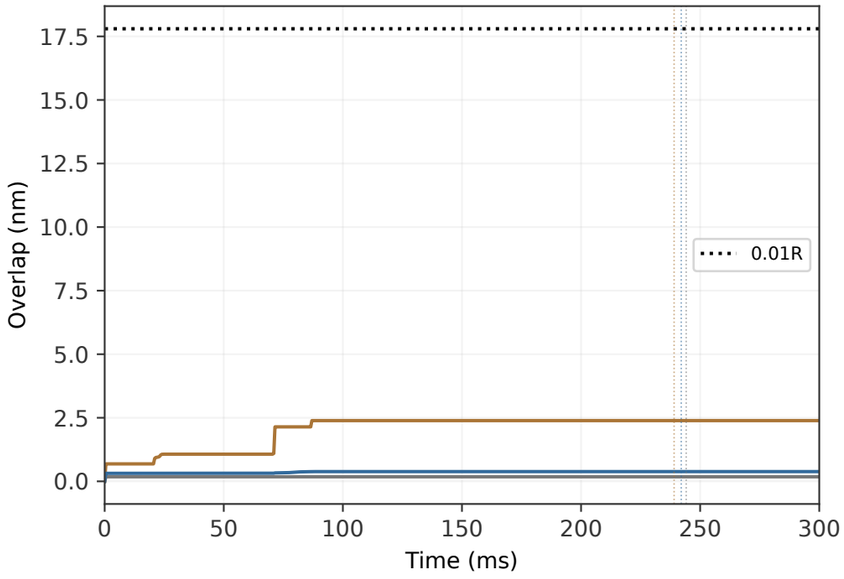
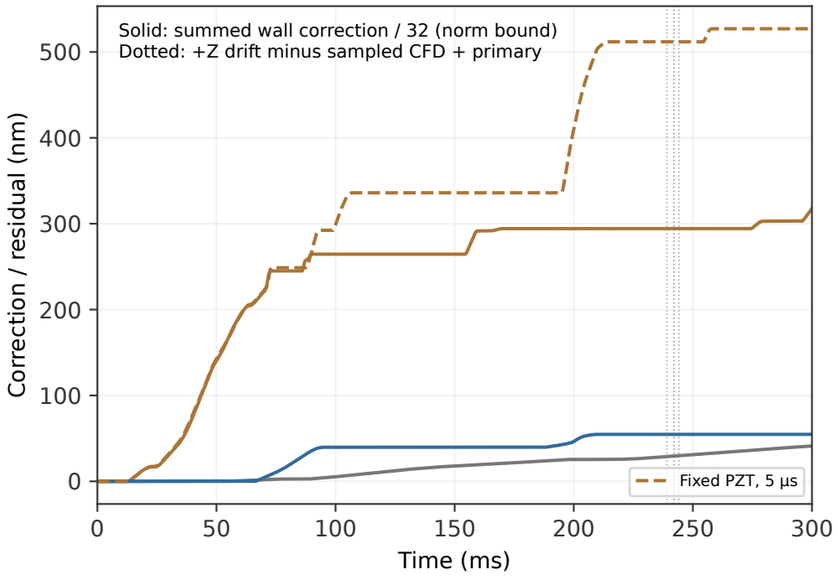
(f) Feedback amplitudes



(g) Feedback phases



(h) Cumulative max bubble overlap



(i) Wall correction and group-drift diagnostic